

PROJETO DE ANÁLISE DE DADOS

Walmart

Sales Analysis

9 perguntas de negócio respondidas
com Python + SQL (MySQL)

10 050

transações

~\$1 209 726

em vendas

100

filiais

Desliza para ver os resultados >

Do CSV ao MySQL

Limpeza e preparação dos dados com Python

1

Carregar

`pd.read_csv()` - 10 050 registos brutos

2

Duplicados

`drop_duplicates()`

3

Nulos

`dropna()` - remover linhas incompletas

4

Tipos

`unit_price`: remover "\$" e converter para float

5

Nova coluna

`total = unit_price x quantity`

6

Carregar em MySQL

`to_sql()` via SQLAlchemy + PyMySQL

RESULTADO

9969 registos limpos, prontos para SQL.

Métodos de pagamento

Nº de transações e unidades vendidas

Question 1: “What are the different payment methods, and how many transactions and items were sold with each method?”

Question 4: “How many items were sold through each payment method?”

Pagamento	Transações	Unidades
Credit card	4 256	9 567
Ewallet	3 881	8 932
Cash	1 832	4 984

Q1 e Q4 usam a mesma query.

```
SELECT
    payment_method,
    COUNT(*) as no_payments,
    SUM(quantity) as no_qty_sold
FROM walmart
GROUP BY payment_method;
```

INSIGHT

Cartão de crédito lidera: 43% das transações.
Dinheiro representa só 18%.

Categoria com melhor avaliação

Por filial

branch	category	avg_rating	ranking
WALM001	Electronic accessories	7,45	1
WALM002	Food and beverages	8,25	1
WALM003	Sports and travel	7,50	1
WALM004	Food and beverages	9,30	1
WALM005	Health and beauty	8,37	1

Amostra: primeiras 6 de 101 linhas (a query devolve todas).

```
SELECT *
FROM
(
    SELECT
        branch,
        category,
        AVG(rating) as avg_rating,
        RANK() OVER(PARTITION BY branch ORDER BY AVG(rating) DESC) as
        ranking
    FROM walmart
    GROUP BY 1, 2
) AS ranking
WHERE ranking = 1;
```

INSIGHT

Food and beverages e Sports and travel lideram em 26 filiais cada;
Home and lifestyle só em 2.

Dia mais movimentado

Por filial, em nº de transações

branch	day_name	no_transactions	ranking
WALM001	Thursday	16	1
WALM002	Thursday	15	1
WALM003	Tuesday	33	1
WALM004	Sunday	14	1
WALM005	Wednesday	19	1
WALM006	Thursday	15	1
WALM007	Friday	12	1
WALM007	Sunday	12	1
WALM008	Tuesday	17	1

Amostra: primeiras 9 de 114 linhas (a query devolve todas).

```
SELECT *
FROM (
    SELECT
        branch,
        DAYNAME(STR_TO_DATE(date, '%d/%m/%Y')) AS day_name,
        COUNT(*) AS no_transactions,
        RANK() OVER (
            PARTITION BY branch
            ORDER BY COUNT(*) DESC
        ) AS ranking
    FROM walmart
    GROUP BY branch, day_name
) AS ranked
WHERE ranking = 1;
```

Rating por cidade e categoria

Média, mínimo e máximo

city	category	min_rating	max_rating	avg_rating
Abilene	Electronic accessories	7,1	8,8	7,97
Abilene	Fashion accessories	4,0	9,0	6,24
Abilene	Food and beverages	6,0	8,9	6,95
Abilene	Health and beauty	9,7	9,7	9,70
Abilene	Home and lifestyle	4,0	9,0	6,10
Alamo	Fashion accessories	3,0	9,0	6,87

Amostra: primeiras 6 de 513 linhas (a query devolve todas).

```
SELECT
    city,
    category,
    MIN(rating) as min_rating,
    MAX(rating) as max_rating,
    AVG(rating) as avg_rating
FROM walmart
GROUP BY city, category;
```

Lucro por categoria

Ranking do maior para o menor

Categoria	Receita (\$)	Lucro (\$)
Fashion accessories	489 481	192 315
Home and lifestyle	489 250	192 214
Electronic accessories	78 175	30 772
Food and beverages	53 471	21 553
Sports and travel	52 498	20 614
Health and beauty	46 851	18 672

```
SELECT
    category,
    SUM(total) AS total_revenue,
    SUM(total * profit_margin) AS profit
FROM walmart
GROUP BY category
ORDER BY profit DESC;
```

INSIGHT

Fashion accessories e Home and lifestyle geram 81% do lucro total.

Pagamento mais usado

Por filial

branch	payment_method	total_trans	ranking
WALM001	Ewallet	45	1
WALM002	Ewallet	37	1
WALM003	Credit card	115	1
WALM004	Ewallet	44	1
WALM005	Ewallet	56	1

Amostra: primeiras 5 de 100 linhas (a query devolve todas).

```
WITH cte
AS
    (SELECT
        branch,
        payment_method,
        COUNT(*) as total_trans,
        RANK() OVER(PARTITION BY branch ORDER BY COUNT(*) DESC) as ranking
    FROM walmart
    GROUP BY branch, payment_method
    )
SELECT *
FROM cte
WHERE ranking = 1;
```

INSIGHT

Ewallet é o nº 1 em 75 filiais, apesar do crédito liderar no total. Dinheiro só em 2.

Q8

"How many transactions occur in each shift (Morning, Afternoon, Evening) across branches?"

Transações por turno

Manhã, tarde e noite

day_time	transactions
Afternoon	4 636
Evening	3 246
Morning	2 087

```
SELECT
    CASE
        WHEN EXTRACT(HOUR FROM time) < 12 THEN 'Morning'
        WHEN EXTRACT(HOUR FROM time) BETWEEN 12 AND 17 THEN
            'Afternoon'
        ELSE 'Evening'
    END AS day_time,
    COUNT(*) AS transactions
FROM walmart
GROUP BY 1;
```

INSIGHT

A tarde concentra 47% das transações.
A manhã é o turno mais fraco.

Quebra de receita 2022 vs 2023

Top 5 filiais com maior queda

branch	last_year_revenue	current_year_revenue	revenue_decrease_ratio
WALM045	1 731,00	647,00	62,62
WALM047	2 581,00	1 069,00	58,58
WALM098	2 446,00	1 030,00	57,89
WALM033	2 099,00	931,00	55,65

```
WITH revenue_2022 AS (  
    SELECT  
        branch,  
        SUM(total) AS revenue  
    FROM walmart  
    WHERE YEAR(STR_TO_DATE(date, '%d/%m/%Y')) = 2022  
    GROUP BY branch  
,  
revenue_2023 AS (  
    SELECT  
        branch,  
        SUM(total) AS revenue  
    FROM walmart  
    WHERE YEAR(STR_TO_DATE(date, '%d/%m/%Y')) = 2023  
    GROUP BY branch  
)  
SELECT  
    r2022.branch,  
    r2022.revenue AS last_year_revenue,  
    r2023.revenue AS current_year_revenue,  
    ROUND(((r2022.revenue - r2023.revenue) / r2022.revenue) * 100, 2) AS  
    revenue_decrease_ratio  
FROM revenue_2022 AS r2022  
JOIN revenue_2023 AS r2023 ON r2022.branch = r2023.branch  
WHERE r2022.revenue > r2023.revenue  
ORDER BY revenue_decrease_ratio DESC  
LIMIT 4;
```

Principais conclusões

- Cartão de crédito lidera o total; Ewallet vence em 75 filiais.
- Moda e Casa geram ~80% do lucro.
- A tarde concentra a maioria das transações.
- Quinta-feira é o dia forte na maioria das filiais.
- Algumas filiais caíram mais de 50% entre 2022 e 2023.

Obrigado!

Stack: Python (pandas, SQLAlchemy) + MySQL

Nuno Cruz

Código no GitHub:

[git@github.com:NunoCruz1/walmart_sales-SQL-Python-.git](https://github.com/NunoCruz1/walmart_sales-SQL-Python-.git)